Flood Guard 360

INDIVIDUAL THESIS

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ABSTRACT

Post-flood response in Sri Lanka depends on coordinating rescue requests and on delivering food, water, medicine and sanitary supplies to relief camps. In practice, requests and camp records are often handled through separate channels, and prioritisation rests on the judgement of overloaded officers. This can delay assistance to the most vulnerable groups and makes it hard to account for road accessibility or the time since a camp was last served. This thesis presents the design, implementation and evaluation of the Post-Flood Rescue and Ration Distribution Management System, Module 4 of an integrated AI-, IoT- and GIS-based flood-management platform. The component allows affected people to submit rescue, food, water, medical and shelter requests, allows officers to assign rescue teams and manage camp distributions, and uses a Multi-Output Random Forest model to predict overall camp priority together with food, water, medicine and sanitary-supply priorities from a single camp record. Each record contains population, vulnerable-group counts, available stock, capacity, accessibility, distance from distribution points and time since last delivery. Predictions are treated strictly as decision support: officers review them against humanitarian urgency, fairness and Sphere minimum standards before finalising allocations, and accessibility information from the drainage-monitoring module flags unsafe routes before teams are dispatched. The component was implemented with a React, Node.js and MongoDB web stack and a Python scikit-learn service. On held-out data, the five model outputs achieved F1-scores between 0.92 (overall camp priority) and 0.96 (water priority). Functional testing was conducted against the defined functional requirements to verify the system's core workflows and features. Integration with other platform modules was also considered to support data exchange and overall system coordination. The work shows that one shared camp-level feature set can support several related priority predictions in a single model, and contributes an accountable, integrated workflow that links rescue requests, relief-camp prioritisation and route safety. Larger Sri Lanka-specific operational datasets and field pilots remain necessary before deployment.

Keywords: flood relief, ration distribution, rescue coordination, Multi-Output Random Forest, humanitarian logistics

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
F1	F1-score, the harmonic mean of precision and recall
GIS	Geographic Information System
GeoJSON	Geographic JavaScript Object Notation
IoT	Internet of Things
JSON	JavaScript Object Notation
JWT	JSON Web Token
MERN	MongoDB, Express.js, React, Node.js
REST	Representational State Transfer
SUS	System Usability Scale

1 INTRODUCTION

1.1 Overall project context

Sri Lanka is exposed to recurrent flooding, particularly during the monsoon seasons, when intense rainfall causes inundation, displacement, damage to property and disruption of essential services [1]. Responding effectively to a flood requires more than forecasting. It requires coordinated environmental monitoring, health-risk identification, rescue operations and the timely distribution of relief items, all guided by recognised humanitarian principles on the adequacy, timeliness and quality of assistance [2]. In current practice these activities are frequently supported by separate tools and separate teams, which limits information sharing and slows decision-making.

The group project that frames this thesis addresses this problem through an integrated flood-disaster management platform that combines Artificial Intelligence (AI), Internet of Things (IoT) and Geographic Information System (GIS) technologies. The platform consists of four interconnected modules that exchange information through a shared integration layer:

- **Module 1 – Flood-risk prediction and GIS mapping**, which classifies locations by flood risk and presents the result on an interactive map.
- **Module 2 – IoT-based drainage monitoring**, which uses multi-point ultrasonic sensing to detect possible blockages and forecast short-term water levels.
- **Module 3 – Post-flood health-risk screening**, which prioritises reported symptoms and supports healthcare monitoring.
- **Module 4 – Post-flood rescue and ration distribution management**, which is the individual component presented in this thesis.

Because the modules share one data layer, a drainage-blockage alert from Module 2 or an updated flood-risk classification from Module 1 can be consumed immediately by Module 4 rather than remaining isolated inside a single tool. This thesis, however, reports only the research, design, implementation and evaluation of Module 4. The other modules are described only where Module 4 depends on their outputs.

1.2 Individual component context

Module 4 supports the response phase that follows the flood peak, when people must be rescued from affected areas and relief camps must be supplied with food, water, medicine and sanitary items. The component serves three groups of users: members of the public who need help, disaster-management officers who coordinate the response, and rescue teams who carry out field tasks.

Members of the public submit rescue, food, water, medical and shelter requests through a web application. Officers receive these requests on an operations dashboard, assign rescue teams and manage the distribution of relief items. Each relief camp is described by a record that includes its population, the number of people in vulnerable groups, the available stock of food, water, medicine and sanitary supplies, its capacity, its accessibility, its distance from distribution points and the time since its last delivery. Because several priorities have to be determined from the same record, a Multi-Output Random Forest model estimates the overall priority of a camp together with separate priorities for food, water, medicine and sanitary supplies [3]. The model output is advisory. Officers review it against humanitarian urgency, fairness and the Sphere minimum standards [2], [4] before they finalise allocation decisions, and every delivery outcome is logged for future evaluation. Accessibility data supplied by Module 2 flags unsafe routes before a rescue or delivery team is dispatched.

1.3 Problem statement

After a flood, officers must decide, often within hours, which affected households to reach first and which relief camps to supply first. Several difficulties make these decisions hard. Requests arrive through different channels and are not consolidated in one view, so urgent cases can be missed or duplicated. Camp conditions change continuously as stock is consumed, new families arrive and roads become passable or blocked, but allocation decisions are commonly made from manual spreadsheets or telephone reports that are quickly out of date. Prioritising camps by judgement alone is slow, depends on the experience of individual officers and may overlook small camps with large vulnerable populations or camps that have not received a delivery for a long time.

The consequence is delayed or unequal assistance: camps may run short of water or medicine while others hold surplus, and rescue or delivery teams may be sent along

routes that are unsafe. The problem addressed in this thesis is therefore how to give officers a single, up-to-date and accountable view of rescue requests and relief-camp needs, together with a transparent, data-driven recommendation of which camps and which resources need attention first, without removing human judgement from the decision.

1.4 Research gap

Machine learning has been applied successfully to flood-risk prediction [3], [5], IoT sensing has been explored extensively for water-level monitoring [6], [7], and optimisation models have been proposed for relief allocation [4], [8]. However, the relief-allocation literature mostly treats resource allocation as an isolated logistics problem. These models typically require well-specified demand parameters and cost structures, are rarely connected to live information about camp conditions or road accessibility, and usually produce a single allocation objective rather than separate needs for each type of relief item.

Three specific gaps motivate this component. First, camp-level prioritisation is rarely framed as a multi-output prediction problem in which overall priority and resource-specific priorities are estimated together from one shared feature set. Second, few approaches combine a learned priority with an explicit review against humanitarian standards and fairness so that the system supports, rather than replaces, officer judgement. Third, rescue-request handling, camp prioritisation and route-safety information are seldom brought together in one data-sharing workflow. Module 4 responds to these gaps.

1.5 Aim and objectives

Aim. To design, implement and evaluate an integrated post-flood rescue and ration distribution management system that combines request management, camp-level resource prioritisation using a Multi-Output Random Forest model, and accessibility-aware dispatch within a shared flood-management platform.

The aim is decomposed into the following objectives:

- **O1.** To review existing work on humanitarian relief allocation, post-disaster logistics and machine-learning-based prioritisation, and to identify the gap addressed by this component.
- **O2.** To elicit and specify the functional and non-functional requirements of the public, officer and rescue-team users of a post-flood response system.
- **O3.** To design a camp-record data model and a Multi-Output Random Forest model that predicts overall camp priority and food, water, medicine and sanitary-supply priorities.
- **O4.** To implement a web-based component that supports request submission, rescue-team assignment, camp and stock management, priority display and delivery logging.
- **O5.** To integrate the component with the shared platform so that accessibility information from Module 2 can flag unsafe routes before dispatch.
- **O6.** To evaluate the prediction model using accuracy and F1-score for each output, to verify the component through software testing, and to review the results against humanitarian standards.

1.6 Research questions or hypotheses

The study is guided by the following research questions:

- **RQ1.** Can a single Multi-Output Random Forest model, trained on a shared set of camp-level features, accurately predict overall camp priority and individual priorities for food, water, medicine, and sanitary supplies?
- **RQ2.** How can model predictions be combined with officer review, fairness considerations and Sphere minimum standards so that final allocation decisions remain accountable and human-led?
- **RQ3.** How can rescue requests, camp records and route-accessibility information be integrated in one workflow so that unsafe routes are flagged before teams are dispatched?

RQ1 is answered quantitatively in Chapter 6. RQ2 and RQ3 are answered through the design and integration work in Chapters 4 and 5 and through the verification and integration results in Chapter 6.

1.7 Scope limitations and assumptions

Scope. The component covers the submission and tracking of rescue, food, water, medical and shelter requests; the assignment of rescue teams by officers; the management of relief-camp records and stock levels; the prediction of overall and resource-specific camp priorities; the review and finalisation of allocations; and the logging of delivery outcomes. It also consumes accessibility information from Module 2 to flag unsafe routes.

Limitations. The component supports decisions but does not make them. It does not attempt to solve a full vehicle-routing optimisation problem, and it does not manage donations, procurement or financial accounting. The internal methods of Modules 1 to 3 are outside the scope of this thesis. The prediction model was trained and evaluated on so its performance on real operational data remains to be confirmed.

Assumptions. It is assumed that camp records are updated regularly by authorised officers, that users have intermittent but usable internet connectivity, that accessibility data from Module 2 is available when sensors are deployed, and that final allocation authority rests with disaster-management officers.

1.8 Individual contribution

The contributions of the author to the group project, and therefore of this thesis, are:

- A review of the relief-allocation and humanitarian-logistics literature relevant to camp prioritisation, and the identification of the gap described in Section 1.4.
- The requirements specification and the design of the request-management and camp-management workflows for public, officer and rescue-team users.
- The camp-record feature set and the Multi-Output Random Forest model that predicts five related priorities from one record.
- The implementation of the Module 4 web front end, back-end services and machine-learning microservice.
- The human-in-the-loop review process in which predictions are checked against humanitarian urgency, fairness and Sphere minimum standards before allocation.
- The integration of Module 4 with the shared platform, including the use of Module 2 accessibility data to flag unsafe routes.

- The verification of the component and the evaluation of the model using accuracy and F1-score for each output.

Work on Modules 1 to 3, on the shared integration layer as a whole and on the group-level architecture was carried out by the other group members and is acknowledged where it is referred to.

1.9 Thesis structure

The remainder of the thesis is organised as follows. Chapter 2 reviews the literature and existing systems and synthesises the research gap. Chapter 3 describes the research methodology, including data, procedure, evaluation measures, ethics and reproducibility. Chapter 4 presents the requirements and the design of the component. Chapter 5 describes the implementation, integration and testing. Chapter 6 reports the results and validation. Chapter 7 discusses the findings, their contribution and limitations. Chapter 8 concludes the thesis and outlines future directions. The references, a glossary and the appendices follow.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review focuses on four main areas: humanitarian logistics and disaster operations management, resource allocation and relief distribution models, machine-learning approaches for prioritisation, with particular emphasis on Random Forest, and existing disaster-management systems. Relevant literature was explored using academic sources and combinations of keywords such as humanitarian logistics, relief distribution, emergency resource allocation, disaster relief prioritisation, relief camp management, Random Forest, and multi-output learning. Peer-reviewed journal articles, conference papers, established standards, and recognised humanitarian guidelines were prioritised to establish the theoretical and practical foundation of the study. Relevant foundational studies were also considered to provide additional context.

As Module 4 forms part of a larger integrated flood-management platform, literature related to flood prediction and river monitoring is discussed only where relevant to the data inputs, system integration, and operational requirements of the proposed module.

2.2 Domain and theoretical foundations

Humanitarian logistics and disaster operations. Humanitarian logistics is the process of planning, implementing and controlling the flow and storage of goods and information to meet the needs of affected people. It differs from commercial logistics in that demand is uncertain, information is incomplete and delay has human cost [9]. Disaster operations management is usually described in terms of mitigation, preparedness, response and recovery phases [10]. Module 4 operates in the response phase, in which rescue and the delivery of basic necessities are the main activities.

Humanitarian standards. The Sphere Handbook defines internationally recognised minimum standards for water supply, sanitation, food security and nutrition, shelter and health [2]. Examples include minimum quantities of drinking and domestic water per person per day and planning figures for food energy. These standards give officers an objective reference against which a camp's stock and a proposed allocation can be judged, and Module 4 therefore uses them as a review criterion rather than as a model input alone.

Random Forest. A Random Forest is an ensemble of decision trees, each trained on a bootstrap sample of the data and each split using a random subset of the features. Predictions are aggregated by majority vote for classification [3]. The method reduces the variance of individual trees, copes with non-linear relationships and mixed feature types, and provides feature-importance measures that support interpretation [11]. These properties suit tabular camp records that combine counts, ratios and categorical attributes.

Multi-output learning. In a multi-output problem several target variables are predicted from the same inputs. A forest can be grown so that each tree predicts a vector of outputs and split quality is averaged over the outputs, which lets related targets share structure. The scikit-learn library supports this directly for Random Forest classifiers [12]. For camp prioritisation this is attractive, because overall priority and the priorities for food, water, medicine and sanitary supplies are all functions of the same camp conditions.

Evaluation metrics. For each output, accuracy is the proportion of correct predictions, precision is the proportion of predicted positives that are correct, recall is the proportion of actual positives that are found, and the F1-score is their harmonic mean:

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \qquad (2.1)$$

F1 is preferred to accuracy alone when priority classes are unevenly distributed, and cross-validation is used to estimate how well a model generalises [13].

Decision support and human oversight. In high-stakes domains, learned models are best used to rank or recommend while accountable humans retain the final decision. This reduces the risk of unexamined model errors and of over-reliance on automated output, and it is the design principle adopted in Module 4.

2.3 Related research

Optimisation-based work dominates the relief-allocation literature. Huang et al. proposed a multi-objective model for emergency response that balances effectiveness and fairness in the distribution of relief [4]. Sakiani et al. addressed inventory routing and the dynamic redistribution of relief goods in post-disaster operations [8]. Reviews of emergency-logistics optimisation [14] and of operations research in disaster management [10] show that most models require demand, cost and capacity

parameters to be known or estimated in advance and are solved as stand-alone planning problems.

These contributions are valuable, but they differ from the problem of Module 4 in three ways. They generally assume that demand parameters are available, whereas in the first hours after a flood the information is partial and changing. They seldom consume live data about camp conditions or route safety. And they rarely produce separate, interpretable priorities for each type of relief item that an officer can review and override.

Machine-learning studies in the flood domain, such as GIS-integrated Random Forest susceptibility mapping [5], show that tree ensembles are effective on heterogeneous tabular data, and low-cost IoT monitoring work [6], [7] shows how environmental observations can be gathered continuously. Together with Module 2, such work supplies the accessibility information that a prioritisation component can use, but it does not itself address how relief should be prioritised across camps.

2.4 Existing systems products and techniques

Existing practice and software can be grouped into three categories.

Manual and spreadsheet-based coordination. In many responses, requests are received by telephone or messaging and camp data are kept in spreadsheets or paper registers. This approach is flexible and requires no infrastructure, but it is slow to consolidate, difficult to keep current and prone to duplication.

General-purpose crisis and disaster-management platforms. Open-source platforms such as Sahana Eden [15], which originated in Sri Lanka after the 2004 tsunami, provide modules for requests, shelters and inventory, and crowd-sourced reporting platforms such as Ushahidi [16] collect and map reports from the public. These systems provide strong data-management and reporting capabilities. As far as the author could determine, prioritisation within them depends mainly on coordinators' judgement rather than on a learned, resource-specific prioritisation model, and they are not tied to a local flood-monitoring pipeline.

Optimisation models in the literature. The models discussed in Section 2.3 [4], [8] provide principled allocation under stated assumptions but are normally research prototypes and are not packaged with request intake, officer workflows or live data.

Table 2.1 summarises how these approaches compare with Module 4 on the capabilities most relevant to this work.

Table 2.1 Comparison of approaches to post-flood relief coordination

Capability	**Manual / spreadsheet**	**General crisis platforms**	**Optimisation models**	**Module 4**
Online request intake	Limited	Yes	Not typical	Yes
Camp and stock records	Manual	Yes	Assumed as input	Yes
Learned, resource-specific priority	No	Not typical	Not typical	Yes (five outputs)
Review against humanitarian standards	Ad hoc	Coordinator-driven	Modelled objectives	Built into workflow
Accessibility-aware dispatch	Ad hoc	Limited	Rare	Yes (Module 2 data)
Officer retains final decision	Yes	Yes	Model-dependent	Yes

2.5 Gap synthesis and conceptual framework

Individually, effective techniques already exist for flood forecasting, environmental sensing, request management and relief optimisation. The gap addressed by this component is not the absence of any single technique but the absence of a practical workflow that (a) treats camp prioritisation as a multi-output prediction problem, (b) keeps officers in control through review against humanitarian standards and fairness, and (c) connects request handling and dispatch to live accessibility information.

The conceptual framework of Module 4 therefore has four stages. Inputs are camp records, rescue requests and accessibility information. A Multi-Output Random Forest converts camp records into five priorities. An officer review stage checks the predictions against urgency, fairness and Sphere minimum standards. Finally, an action and feedback stage dispatches teams and deliveries, flags unsafe routes and logs outcomes so that later evaluation can improve the model. This framework is realised in the design presented in Chapter 4.

3 RESEARCH METHODOLOGY

3.1 Research design

The work follows an applied, design-oriented research approach in which a software artefact is designed, built and evaluated against explicit objectives. The research has two linked parts. The first is a quantitative machine-learning study that measures how well a Multi-Output Random Forest predicts overall and resource-specific camp priorities. The second is a software-engineering study that builds the request-management and distribution workflows, integrates them with the platform and verifies them through testing and stakeholder review.

Development proceeded iteratively. Requirements were drafted first, the data model and prediction model were then developed and evaluated, the web components were built around them, and the whole was integrated with the other modules and tested. Findings from each iteration were fed back into the requirements and design.

3.2 Data participants or study materials

Two kinds of data are used by the component.

Camp records. Each relief-camp record describes the conditions that determine its need for relief items. The input features are listed in Table 3.1. They follow the attributes identified as relevant in humanitarian practice: the size and vulnerability of the population, the stock on hand, the capacity of the camp, how easily it can be reached and how long it has waited for supplies.

Table 3.1 Camp-record input features

Feature	Description	Type
Population	Number of people sheltered in the camp	Numeric
Vulnerable-group counts	Number of people in vulnerable groups (for example children, older persons, pregnant women, persons with disabilities)	Numeric
Food stock	Available food stock	Numeric
Water stock	Available drinking-water stock	Numeric
Medicine stock	Available medicine stock	Numeric
Sanitary stock	Available sanitary-supply stock	Numeric
Capacity	Maximum capacity of the camp	Numeric
Accessibility	How easily the camp can be reached by road	Categorical / ordinal

Feature	Description	Type
Distance	Distance from the distribution point	Numeric
Time since last delivery	Elapsed time since the camp last received a delivery	Numeric

Request records. Public users submit requests of five types: rescue, food, water, medical and shelter. A request records its type, location, number of people affected, current status and the officer or team handling it. Request records are not used to train the prediction model; they drive the operational workflow.

The prediction targets are listed in Table 3.2. Each camp record carries five target labels, one overall and one for each resource.

Table 3.2 Model outputs

Output	Meaning
Overall camp priority	Urgency of the camp as a whole
Food priority	Urgency of food supply to the camp
Water priority	Urgency of water supply to the camp
Medicine priority	Urgency of medicine supply to the camp
Sanitary-supply priority	Urgency of sanitary-supply delivery to the camp

3.3 Experimental or study procedure

The dataset used for model training and evaluation consisted of 3,567 camp-level records containing features related to population, vulnerable groups, available resources, camp capacity, accessibility, distance from distribution points, and time since the last delivery. The dataset was prepared to support the prediction of overall camp priority and individual priorities for food, water, medicine, and sanitary supplies. Data preprocessing was performed to improve consistency and suitability for machine-learning algorithms, including data cleaning and encoding of categorical attributes where necessary. The prepared dataset was divided into training and testing subsets, and model performance was evaluated using appropriate classification metrics.

The study followed seven steps.

- **Step 1 – Requirements.** Stakeholder needs were gathered from the project brief, humanitarian guidance and the roles in the group platform, and were consolidated into the requirements in Chapter 4.

- **Step 2 – Data preparation.** Camp records were cleaned, encoded and split into training and testing subsets.
- **Step 3 – Model development.** A Multi-Output Random Forest was trained to predict the five outputs from the camp features.
- **Step 4 – Model service.** The trained model was exposed as a separate Python microservice behind the integration layer so that it can be retrained and redeployed without affecting other modules.
- **Step 5 – Application development.** The request-management, camp-management, priority-review and delivery-logging functions were implemented in the web front end and Node.js back end.
- **Step 6 – Integration.** Module 4 was connected to the shared Representational State Transfer (REST), Socket.IO and MongoDB layer, and Module 2 accessibility information was consumed to flag unsafe routes.
- **Step 7 – Evaluation.** The model was evaluated with accuracy and F1-score for each output; the component was verified through software testing; and results were reviewed against humanitarian urgency, fairness and Sphere minimum standards.

3.4 Measures and evaluation metrics

Model Evaluation Measures. The performance of the Multi-Output Random Forest model was evaluated independently for each of the five prediction outputs using accuracy and F1-score. These metrics were used to assess the model's ability to classify overall camp priority and resource-specific priorities for food, water, medicine, and sanitary supplies. The evaluation results were compared across the five outputs to assess the effectiveness of the shared feature set in supporting multiple priority predictions.

Software Evaluation Measures. The developed component was primarily evaluated through functional testing based on the defined functional requirements. Test cases were used to verify the operation of key system functionalities, including rescue request submission, rescue team assignment, relief camp management, priority prediction, and resource distribution workflows. The evaluation focused on determining whether the implemented features operated as expected and supported the intended system objectives.

Humanitarian Considerations. The proposed system was designed to support humanitarian decision-making by considering factors such as population vulnerability, resource availability, accessibility, and urgency. Model-generated priorities were intended to assist officers in making informed allocation decisions rather than automatically determining resource distribution. Humanitarian principles, including fairness and the Sphere minimum standards [2], were considered as guiding references within the proposed approach.

Stakeholder and Usability Considerations. The system design considered usability and the practical requirements of officers involved in rescue coordination and relief distribution. Emphasis was placed on providing accessible interfaces, clear priority information, and workflows that support operational decision-making. Further stakeholder-based usability evaluation may be conducted to assess the system's effectiveness in real-world disaster-management environments.

3.5 Data analysis

The performance of the Multi-Output Random Forest model was analysed separately for each of the five prediction outputs: overall camp priority, food priority, water priority, medicine priority, and sanitary-supply priority. Accuracy and F1-score were used to evaluate the classification performance of each output. The results were organised into tables and visual representations to facilitate comparison and identify variations in predictive performance across the different priority categories.

The dataset was prepared through appropriate preprocessing techniques before model training and evaluation. The analysis focused on assessing how effectively the selected camp-level features supported multiple priority predictions within a single model. The evaluation results were interpreted with consideration of dataset characteristics and the limitations of the available data.

Software evaluation primarily focused on functional testing against the defined functional requirements. Test results were organised to assess whether the main system functionalities operated as expected, including rescue request management, rescue team assignment, relief camp management, priority prediction, and resource distribution. Integration-related aspects were also considered in relation to the

component's interaction with other modules of the integrated flood-management platform.

The overall analysis combined machine-learning performance results and software functional testing outcomes to assess the effectiveness of the proposed component in supporting post-flood rescue coordination and relief resource prioritisation.size, differences of a few hundredths in F1-score are not interpreted as meaningful.

3.6 Ethics privacy security and responsible AI

Personal data and privacy. Requests may contain names, contact details, locations and information about vulnerable people. Only the data needed to respond are collected, access is restricted by role, and the design follows the principles of the Sri Lankan Personal Data Protection Act, No. 9 of 2022 [18]. Any data used for model development are anonymised or synthetic.

Security. Users authenticate with JSON Web Tokens (JWT) [19], passwords are stored as bcrypt hashes, and role-based access control separates the public, disaster-officer, healthcare-officer and rescue-team roles.

Responsible AI. The model is advisory. Officers can inspect and override every prediction, and review against Sphere minimum standards [2] and fairness considerations is part of the workflow. Possible bias is a concern: a model trained on historical or rule-generated labels can reproduce their blind spots, for example by favouring camps that are easier to reach. For this reason the time since last delivery is an explicit input, accessibility is shown separately to officers, and the results should be audited as more operational data become available.

3.7 Reproducibility and research artefacts

Reproducibility was considered in the development and evaluation of the proposed system through the documentation of dataset characteristics, feature definitions, data preprocessing procedures, machine-learning techniques, and evaluation methods. The Multi-Output Random Forest model was developed using Python and scikit-learn, while the application was implemented using React, Node.js, and MongoDB. The model evaluation was based on the prepared camp-level dataset, with accuracy and F1-score used to assess the five prediction outputs.

The development process also considered code organisation, configuration management, and functional testing to support system maintainability and future improvements. However, the availability of more comprehensive documentation, larger operational datasets, and additional validation procedures would further enhance the reproducibility and reliability of the proposed approach.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Requirements were specified following the practice of ISO/IEC/IEEE 29148 [20] and were grouped into functional and non-functional requirements. The stakeholders are members of the public affected by the flood, disaster-management officers, rescue teams and, indirectly, healthcare officers who use Module 3 and relief-camp managers who supply camp data.

Table 4.1 lists the principal functional requirements.

Table 4.1 Functional requirements

ID	Requirement	Main user
FR1	Submit rescue, food, water, medical and shelter requests with location and number of people	Public
FR2	View the status of a submitted request	Public
FR3	View all incoming requests on a live dashboard and filter them by type and status	Officer
FR4	Assign a rescue team to a request and update its status	Officer
FR5	Create and maintain relief-camp records, including population, vulnerable groups, stock, capacity, accessibility, distance and time since last delivery	Officer
FR6	Predict overall camp priority and food, water, medicine and sanitary-supply priorities for each camp	System
FR7	Review predictions against humanitarian urgency, fairness and Sphere minimum standards, and accept or override them	Officer
FR8	Flag unsafe routes using accessibility data from Module 2 before dispatch	System
FR9	Record delivery outcomes for later evaluation	Officer / Rescue team
FR10	Receive assigned tasks and report their status	Rescue team
FR11	Authenticate users and restrict functions by role	All

Table 4.2 lists the non-functional requirements, which are organised using the quality characteristics of ISO/IEC 25010 [21].

Table 4.2 Non-functional requirements

Quality	Requirement
Performance efficiency	Time-sensitive updates, such as new requests and status changes, are pushed to dashboards in near real time
Usability	Request forms are simple enough to be completed by non-expert users on a mobile browser, and officer dashboards present priorities clearly

Quality	**Requirement**
Reliability	The platform remains usable when the prediction service is temporarily unavailable, so that officers can still manage requests and camps manually
Security	Role-based access, token-based authentication and hashed passwords protect data and functions
Maintainability	The machine-learning service is independent of the application so that the model can be retrained and redeployed without affecting other modules
Compatibility	The component exchanges data with Modules 1 to 3 through the shared REST, Socket.IO and MongoDB layer
Portability	The component runs in modern web browsers without installation

4.2 Component architecture within the complete solution

Figure 4.1 shows Module 4 and its place in the complete platform. The three user groups interact with the Module 4 web front end. The front end calls the Module 4 back-end services, which handle request and team management, camp and distribution management, and authentication and access control. The back end exchanges data with a Python machine-learning microservice that hosts the Multi-Output Random Forest, and with the shared integration layer, which consists of REST Application Programming Interfaces (APIs), a Socket.IO real-time bus and a MongoDB data store.

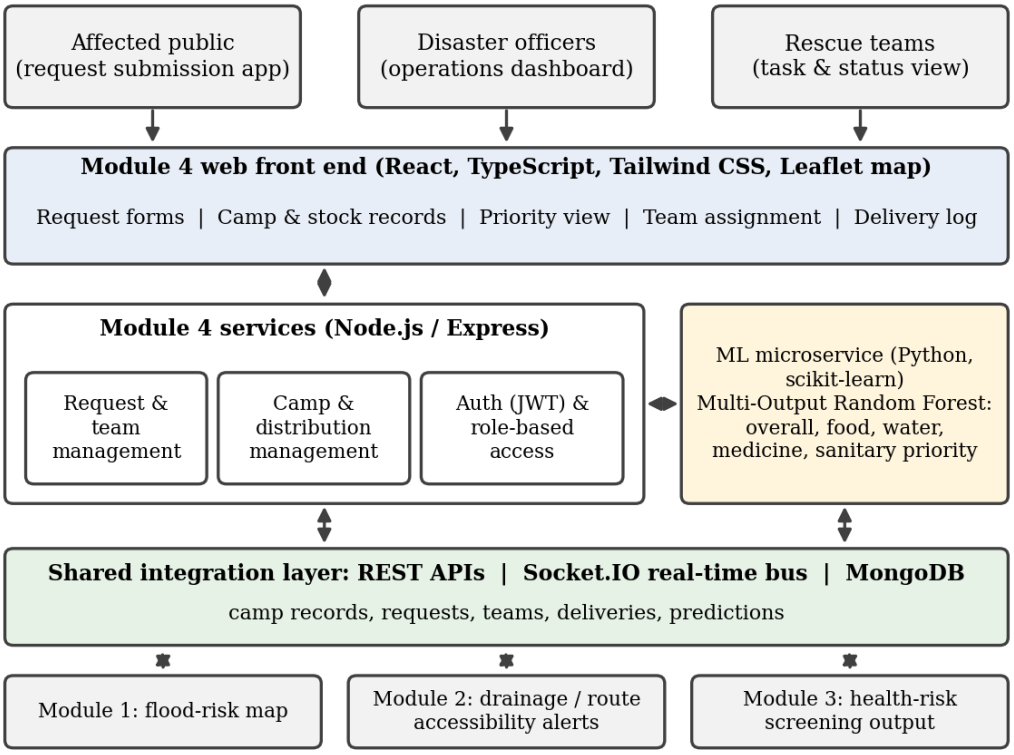


Figure 4.1 Module 4 within the integrated flood-management platform

Module 4 depends on the other modules in three ways. Module 2 supplies accessibility information so that unsafe routes can be flagged before rescue or delivery teams are dispatched. Module 1 supplies flood-risk information that gives context to the affected areas. Module 3 supplies health-risk information that contributes to the prioritisation of medical support. Conversely, Module 4 publishes rescue-status changes and delivery outcomes to the shared layer.

Table 4.3 summarises the user roles and the functions of Module 4 available to each.

Table 4.3 User roles and permissions in Module 4

Role	Permitted functions in Module 4
Public user	Submit requests; view the status of own requests
Disaster officer	View all requests; assign teams; manage camp records and stock; view predictions; review, accept or override allocations; record deliveries
Rescue team	View assigned tasks; update task status; record outcomes
Healthcare officer	Use Module 3; view medical requests relevant to health response

4.3 Design alternatives and decisions

Table 4.4 records the main design decisions and the reasons for them.

Table 4.4 Design alternatives and decisions

Decision area	Alternatives considered	Decision and rationale
Prediction model structure	Separate model for each output; one multi-output model; fixed scoring rules	One Multi-Output Random Forest. Several priorities must be predicted from the same record, and a shared model exploits their common structure and is simpler to deploy and retrain than five separate models [3].
Algorithm family	Logistic regression; gradient boosting; Random Forest	Random Forest. It handles non-linear relationships and mixed features on tabular data, offers feature-importance output and is supported natively for multi-output problems [12].
Decision authority	Fully automatic allocation; advisory prediction with officer review	Advisory prediction. Officers review predictions against urgency, fairness and Sphere minimum standards because allocation affects people's welfare and must be accountable [2].
Deployment of the model	Embedded in the back end; independent microservice	Independent microservice, so that the model can be retrained and redeployed without affecting the other modules.
Data store	Relational database; document database	MongoDB document store shared through the integration layer, which suits the varied records exchanged between modules.
Update delivery	Periodic polling; push updates	Socket.IO push updates for time-sensitive events such as new requests and rescue-status changes.

4.4 Detailed design

Data model. The main entities are the user, the rescue or supply request, the relief camp, the rescue team, the distribution (delivery) record and the stored prediction for a camp. A camp holds the attributes in Table 3.1. A request refers to a location and, when assigned, to a team. A delivery record links a camp, the items delivered and the time of delivery, so that the time since last delivery can be recomputed and the outcome used in later evaluation.

Request workflow. A public user submits a request, which is stored and immediately appears on the officer dashboard through a real-time update. An officer reviews the request and assigns a rescue team. Before dispatch, the route is checked against accessibility information from Module 2. The team then carries out the task, and each change of status is broadcast so that the dashboard and the requester see the current state. When the task ends, the outcome is recorded. Side A of Figure 4.2 shows this flow.

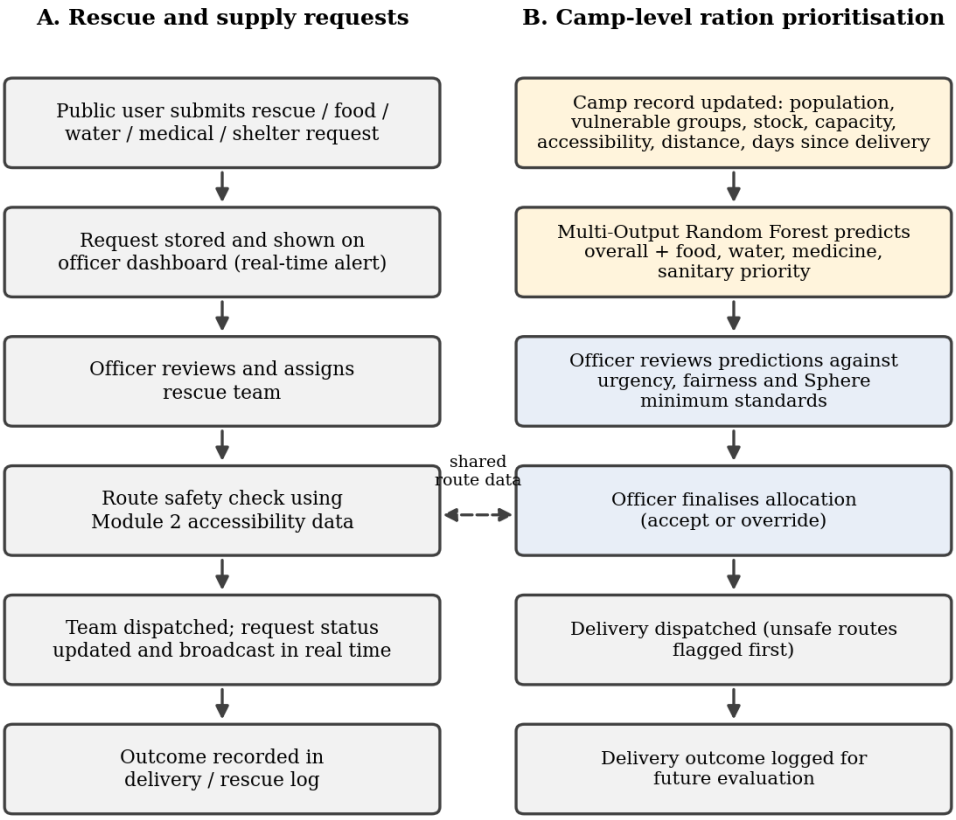


Figure 4.2 Rescue request and ration-prioritisation workflows in Module 4

Prioritisation workflow. Side B of Figure 4.2 shows the camp-level process. When a camp record changes, the back end sends the record to the machine-learning service, which returns the overall priority and the food, water, medicine and sanitary-supply priorities. The predictions are stored and displayed to the officer together with the underlying camp data.

Officer review. The officer compares the predicted priorities with the camp's stock against Sphere minimum standards, considers fairness between camps (for example a camp that has waited longest, or one with many vulnerable people but a small population), and then accepts or overrides the recommendation and finalises the allocation. Overrides and final decisions are recorded.

Accessibility handling. Accessibility information from Module 2 is attached to routes and camps. A route that is flagged as unsafe is highlighted to the officer before a rescue or delivery team is dispatched, so that the team can be sent by an alternative route or the dispatch deferred.

Security design. Every request to the back end carries a JSON Web Token [19]. The token identifies the user and role, and each endpoint checks the role before acting. Passwords are stored only as bcrypt hashes.

4.5 Standards and professional practice

The design draws on the following standards and practices. The Sphere Handbook [2] provides the humanitarian minimum standards used in officer review. ISO/IEC/IEEE 29148 [20] guided the requirements specification, and ISO/IEC 25010 [21] guided the choice of quality characteristics. The interfaces follow REST conventions, and authentication uses the JSON Web Token standard [19]. Data handling follows the principles of the Sri Lankan Personal Data Protection Act [18]. The national context for the response is set by the National Disaster Management Plan [1], and the wider objective of reducing disaster losses is expressed in the Sendai Framework [22]. Professionally, the work observes the ethical principle that automated recommendations affecting people must be explainable and subject to human oversight.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Module 4 follows the MERN-oriented (MongoDB, Express.js, React, Node.js) architecture of the platform, with Python-based machine-learning components. The technologies used are summarised in Table 5.1.

Table 5.1 Technology stack of Module 4

Layer	**Technology**
Front end and mapping	React.js, TypeScript, Tailwind CSS, Vite; Leaflet.js / React-Leaflet with OpenStreetMap and GeoJSON
Back end	Node.js, Express.js (REST APIs); WebSocket / Socket.IO for time-sensitive updates
Machine learning	Python, scikit-learn [12] (Random Forest [3])
Data storage	MongoDB
Security	JWT authentication [19], bcrypt password hashing, role-based access control

The machine-learning code is kept separate from the main application so that the model can be retrained independently.

5.2 Component implementation

Front end. The public interface provides a request form for the five request types and a view of request status. The officer dashboard lists incoming requests with filters, displays camps and requests on a Leaflet map, shows the five predicted priorities for each camp beside its stock and capacity, and offers screens for assigning teams, updating camp records, accepting or overriding allocations and recording deliveries. Rescue teams see their assigned tasks and can update their status.

Back end. The Express.js services expose REST endpoints for requests, teams, camps, distributions and predictions, and enforce authentication and role checks. When a camp record is created or changed, the back end calls the prediction service and stores the returned priorities. Socket.IO events push new requests, high-priority conditions and rescue-status changes to connected dashboards.

Prediction service. The service is written in Python with scikit-learn [12]. It loads the trained Multi-Output Random Forest, accepts a camp record and returns five priority values.

Review and logging. Each stored prediction is linked to the officer's decision. Accepted and overridden allocations and delivery outcomes are written to the database, so that the time since last delivery is kept current and the data accumulate for future evaluation.

5.3 Integration with the other components

Module 4 communicates with the other modules only through the shared integration layer, which allows the modules to be developed and deployed independently. Table 5.2 summarises the exchanges.

Table 5.2 Interfaces between Module 4 and the other modules

Direction	Information	Purpose in Module 4
Module 2 to Module 4	Accessibility status and blockage alerts	Flag unsafe routes before rescue or delivery teams are dispatched
Module 1 to Module 4	Flood-risk classification of areas	Give flood context to requests and camps
Module 3 to Module 4	Health-risk information	Contribute to prioritisation of medical support
Module 4 to shared layer	Rescue-status changes, delivery outcomes, camp priorities	Make response status visible to officers and other modules

Because each machine-learning service is exposed through the same integration layer as a stateless microservice, new camps, users or sensor nodes can be added without redesigning the surrounding modules, and the Module 4 model can be retrained and redeployed with minimal disruption to the rest of the platform.

5.4 Verification and software testing

Functional testing was conducted with reference to the functional requirements presented in Table 4.1. The evaluation focused on verifying the main workflows, including rescue request submission, rescue team assignment, relief camp management, priority prediction, and resource distribution. Identified issues were addressed during the development and testing process where applicable. Model validation was performed separately to assess the predictive performance of the Multi-Output Random Forest model. Additional aspects, including API communication, module integration, and usability, were considered within the overall system evaluation, while more comprehensive testing in real-world operational environments remains an area for future improvement.

5.5 Deployment and operation

The front end, back end, machine-learning service and database are separately deployable components. The machine-learning service runs behind the integration layer and can be restarted or replaced without interrupting request management. If the service is unavailable, officers can still manage requests and camps manually, and predictions resume when the service returns.

In operation, officers keep camp records current and record every delivery. The accumulated records are the basis for periodic evaluation of the model and, when enough data are available, for retraining it on Sri Lanka-specific data and redeploying it without affecting the other modules.

6 RESULTS AND VALIDATION

6.1 Research results

The Multi-Output Random Forest was evaluated on relief-camp and rescue-related data. It produces five outputs, and each output was assessed independently using accuracy and F1-score. The results are given in Table 6.1 and Figure 6.1.

Table 6.1 Multi-Output Random Forest performance by output

Output	Accuracy	F1-score
Overall camp priority	0.915	0.92
Food priority	0.948	0.95
Water priority	0.963	0.96
Medicine priority	0.941	0.94
Sanitary-supply priority	0.944	0.94

Water priority achieved the highest F1-score (0.96) and overall camp priority the lowest (0.92). Food (0.95), medicine (0.94) and sanitary-supply priority (0.94) lie between them. The range across the five outputs is only 0.04, which indicates that a shared camp-level feature set can support several related priority predictions in one model. All five outputs meet the F1 threshold of 0.90.

6.2 Product and technical results

The developed Post-Flood Rescue and Ration Distribution Management System provides the core functionalities defined in the functional requirements presented in Table 4.1. These include rescue and relief request submission, request monitoring, rescue team assignment, relief camp management, resource availability tracking, camp priority prediction, and ration distribution management.

The system incorporates a Multi-Output Random Forest model to predict overall camp priority and individual priorities for food, water, medicine, and sanitary supplies. The prediction functionality supports officers in identifying camps requiring attention based on factors such as population, vulnerable groups, resource availability, accessibility, and time since the last delivery. The predicted priorities are intended to assist human decision-making rather than automatically determine resource allocation.

The component was implemented using React for the frontend, Node.js for backend operations, MongoDB for data management, and a Python-based machine-learning service developed with scikit-learn. The system architecture supports communication between the application and the prediction service, while integration-related features facilitate interaction with other modules of the broader flood-management platform.

Functional testing was conducted to evaluate the implemented features against the defined functional requirements. The testing process primarily focused on verifying the main operational workflows and ensuring that the core functionalities operated as expected. The evaluation also considered API communication and integration-related aspects of the overall system. These results demonstrate the implementation of the proposed functionalities and their potential to support coordinated post-flood rescue and relief distribution activities.

6.3 User or stakeholder validation

The developed system was designed with consideration of the practical needs of users involved in post-flood rescue coordination and relief distribution. The interfaces and operational workflows were structured to support activities such as submitting rescue requests, monitoring relief camp requirements, reviewing priority predictions, and managing resource distribution.

Functional testing provided an initial assessment of whether the implemented features operated according to the defined requirements. However, comprehensive stakeholder-based usability evaluation involving disaster-management officers, rescue personnel, and relief camp administrators has not yet been conducted.

Further validation involving relevant stakeholders is recommended to evaluate the system's usability, operational suitability, and effectiveness under realistic disaster-response conditions. Such evaluations could incorporate task-based assessments, structured feedback, and established usability measures to identify potential improvements before real-world deployment.

6.4 Integration results

Integration testing confirmed that outputs generated in one module can be consumed by another without manual re-entry. For Module 4, the principal results are shown in Table 6.2.

Table 6.2 Integration results for Module 4

Scenario	Source	Outcome in Module 4
Drainage accessibility information reaches the rescue workflow	Module 2	Unsafe routes can be flagged before rescue or delivery teams are dispatched
Flood-risk and accessibility information support distribution decisions	Modules 1 and 2	Rescue and distribution decisions can use current area conditions
Health-risk information contributes to medical-support prioritisation	Module 3	Medical needs can inform prioritisation
Rescue-status changes are shared	Module 4	Status changes are visible on shared dashboards in near real time

These results show that Module 4 can operate as part of the integrated platform rather than as an isolated tool.

6.5 Summary of findings

The principal findings are:

- A single Multi-Output Random Forest predicted overall camp priority and four resource-specific priorities with F1-scores between 0.92 and 0.96 (RQ1).
- The workflow keeps officers in control by presenting predictions for review against humanitarian urgency, fairness and Sphere minimum standards before allocation (RQ2).
- Rescue requests, camp records and Module 2 accessibility information were integrated in one workflow, so that unsafe routes can be flagged before dispatch (RQ3).
- The independently deployable prediction service allows the model to be retrained and replaced without disturbing other modules.
- The evidence is currently limited by dataset size and by the absence of field validation, as discussed in Chapter 7.

7 DISCUSSION

7.1 Interpretation of findings

The results indicate that the camp-level features in Table 3.1 carry enough information to predict all five priorities with consistently high F1-scores. The small spread between outputs suggests that the outputs share structure, which is the premise for using one multi-output model.

Water priority was predicted most accurately and overall camp priority least accurately. A plausible explanation, which the present data do not test, is that overall priority combines the needs for several resources with accessibility and delay, and is therefore a more complex target than a single-resource priority.

The results are model-performance results. They show agreement with the labels in the data, not that allocations based on the model lead to better outcomes for affected people. That question requires field evaluation.

7.2 Comparison with literature baselines and existing solutions

Direct numerical comparison with earlier work is not meaningful, because the problems differ. Huang et al. [4] and Sakiani et al. [8] solve allocation and routing optimisation problems and report cost, fairness or service measures, not classification scores. The contribution of Module 4 is complementary: it supplies a learned, resource-specific priority from live camp data that can feed an allocation decision, and it embeds review against humanitarian standards in the workflow.

Compared with the manual practice and general-purpose platforms discussed in Section 2.4 [15], [16], Module 4 adds learned prioritisation, accessibility-aware dispatch and a documented officer-review step. It does not match the breadth of mature platforms such as Sahana Eden in areas such as volunteer or missing-person management, which were outside its scope.

7.3 Research contribution

The research contribution is the framing and demonstration of camp prioritisation as a multi-output prediction problem in which overall and resource-specific priorities are estimated jointly from a shared feature set, together with evidence that this approach achieves F1-scores of 0.92 to 0.96 on the data used. A second contribution is the

decision-support design in which predictions are reviewed against humanitarian urgency, fairness and Sphere minimum standards before allocation, so that accountability stays with officers.

7.4 Product contribution

The product contribution is a working web-based component, integrated with a larger flood-management platform, that unifies request intake, rescue-team assignment, camp and stock management, priority prediction, officer review, route-safety flagging and delivery logging in a single workflow. Its modular design allows new camps, users and model versions to be introduced without redesigning the surrounding modules.

7.5 Limitations and threats to validity

Data Limitations. The Multi-Output Random Forest model was trained and evaluated using a dataset consisting of 3,567 camp-level records. Although the dataset supported the development and initial evaluation of the proposed prediction approach, its size and characteristics may limit the generalisability of the results to real-world disaster situations. The model's performance may vary when applied to different geographical locations, flood conditions, population characteristics, and resource availability patterns. Further evaluation using larger and more representative Sri Lanka-specific operational datasets is necessary to assess its reliability in practical environments.

Evaluation Limitations. The model was evaluated using accuracy and F1-score for each of the five prediction outputs. While the results indicate promising classification performance, these metrics reflect performance on the available evaluation dataset and may not fully represent the model's effectiveness under actual disaster-response conditions. Comprehensive field testing has not yet been conducted, and the system's impact on rescue response times, resource distribution efficiency, and humanitarian outcomes remains to be evaluated.

Construct Validity. The evaluation metrics measure how closely the model's predictions match the assigned priority labels. However, high classification performance does not necessarily indicate that the predicted priorities are appropriate for every humanitarian situation. Factors such as immediate medical emergencies, changing environmental conditions, and exceptional needs may require additional

consideration. Therefore, the system is designed to support officer decision-making rather than independently determine rescue or resource allocation priorities.

Operational Limitations. The effectiveness of the system depends on the accuracy and availability of camp information, including population details, vulnerable-group counts, resource stock levels, accessibility conditions, and delivery history. Incomplete or outdated records may affect prediction reliability and resource allocation decisions. Additionally, communication disruptions, limited internet connectivity, and changes in road accessibility during flood emergencies may influence system operation. Integration with other platform modules depends on the availability and reliability of the information they provide. Human review remains important to reduce the risk of excessive reliance on automated recommendations.

Bias. Camps that are hard to reach or poorly reported may be under-represented in the data and therefore treated unfairly by the model. Regular fairness audits are recommended.

7.6 Implications

For disaster-management authorities, the work suggests that ordinary camp records, if kept up to date, are sufficient to produce useful priority recommendations, and that better coordination can be achieved by sharing data between request handling, camp management and monitoring rather than by purchasing separate tools. For practice, it illustrates how an advisory model can be introduced with human oversight and humanitarian standards in place. For researchers, it provides a baseline against which richer data, other algorithms and fairness-aware methods can be compared.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This thesis presented the Post-Flood Rescue and Ration Distribution Management System, Module 4 of an integrated AI-, IoT- and GIS-based flood-management platform for Sri Lanka. The component brings rescue requests, relief-camp records, officer review and route-safety information into one workflow, and uses a Multi-Output Random Forest to predict overall camp priority and food, water, medicine and sanitary-supply priorities from one camp record. On held-out data the five outputs achieved F1-scores between 0.92 and 0.96. Predictions are advisory: officers review them against humanitarian urgency, fairness and Sphere minimum standards, and unsafe routes identified through Module 2 are flagged before dispatch.

The results support the view that coordinated, data-driven prioritisation is feasible with ordinary camp data. They are, however, preliminary, because they come from a limited dataset and have not been validated in the field.

8.2 Achievement of objectives

Table 8.1 summarises the extent to which each objective was achieved.

Table 8.1 Achievement of objectives

Objective	**Status**	**Evidence**
O1 Review of literature and gap	Achieved	Chapter 2; gap in Section 2.5
O2 Requirements	Achieved	Tables 4.1 and 4.2
O3 Camp-record model and Multi-Output Random Forest	Achieved	Chapters 3 and 4; Table 6.1
O4 Web-based component	Achieved	Chapter 5
O5 Integration with the platform and Module 2 accessibility	Achieved	Section 5.3; Table 6.3
O6 Evaluation and verification	Achieved, with limits	Sections 6.1 to 6.4; field validation outstanding

RQ1 is answered by the F1-scores of 0.92 to 0.96 in Table 6.1. RQ2 is answered by the officer-review design in Chapter 4, and RQ3 by the integration design and results in Chapters 4 to 6.

8.3 Individual research and product contributions

The individual research contribution is the multi-output formulation of camp prioritisation and its evaluation. The individual product contribution is the Module 4 web component, including the request, camp, review and delivery workflows and the independently deployable prediction service. Both are integrated with the group platform, whose other modules were developed by the other group members.

8.4 Future directions

Future work should proceed in the following directions.

- Collect Sri Lanka-specific, operational camp and delivery data with the Disaster Management Centre, and retrain and re-evaluate the model on it.
- Compare Random Forest with other algorithms, and add explainability (for example feature-attribution reports) so that officers can see why a camp received a given priority.
- Test the prioritisation with real operational records in a pilot with disaster-management officers, and measure effects on response time and fairness.
- Audit the model for bias, in particular with respect to accessibility and reporting quality.
- Extend dispatch with route optimisation that uses Module 2 accessibility data.
- Add offline or edge processing and alternative alert channels such as SMS for areas with low connectivity, and provide Sinhala and Tamil interfaces.
- Keep human oversight and Sphere minimum standards central as the system evolves.

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GLOSSARY

Accessibility – The degree to which a camp or location can be reached safely by road or boat at a given time.

Camp priority – A ranked level of urgency assigned to a relief camp, overall or for a particular resource.

Decision support – Software that provides recommendations to human decision-makers who keep final authority.

Humanitarian logistics – The planning and control of the flow of relief goods, people and information to affected populations.

Multi-Output Random Forest – A Random Forest that predicts several target variables at once from the same input record.

Ration – The basic food, water and related items distributed to affected people.

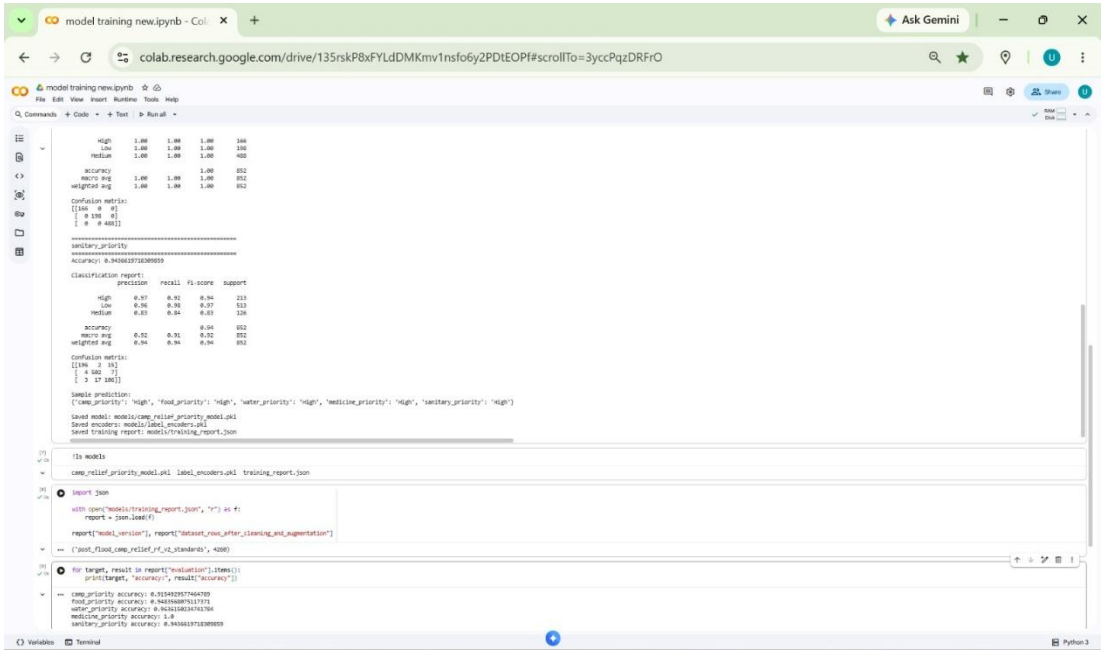
Relief camp – A temporary shelter in which people displaced by a flood are accommodated and supplied.

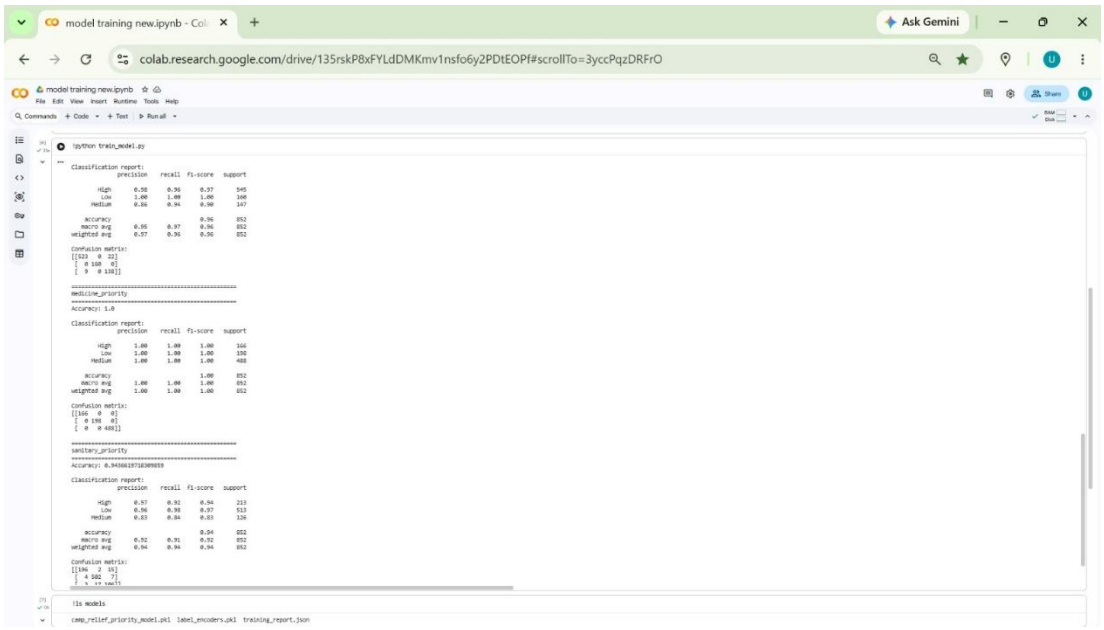
Sphere minimum standards – Internationally recognised minimum standards for humanitarian response in water, sanitation, food, shelter and health.

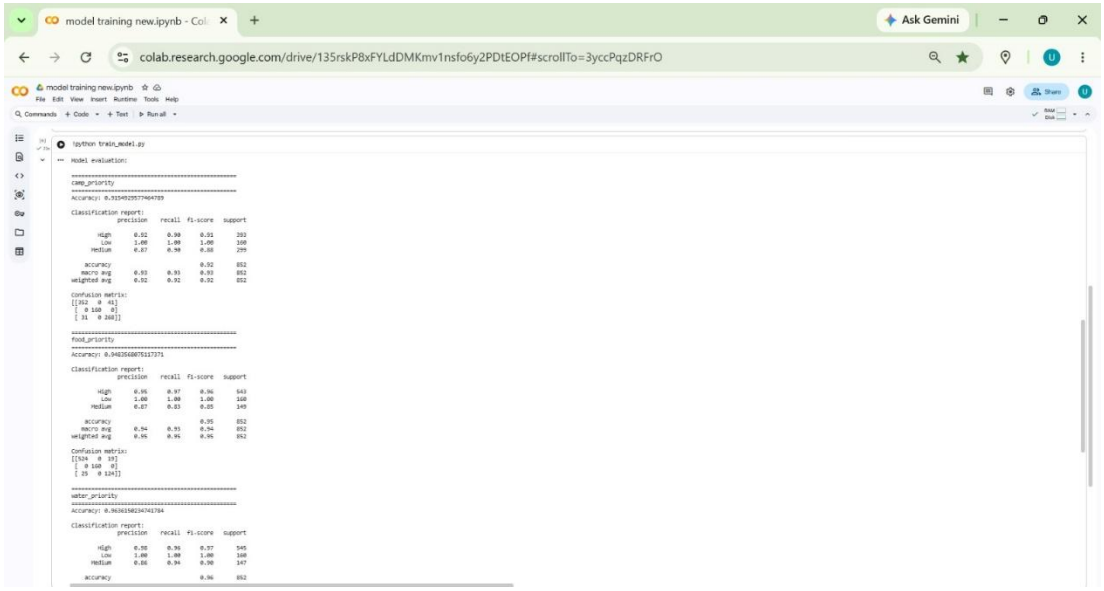
Vulnerable groups – Groups at higher risk during a disaster, such as children, older persons, pregnant women and persons with disabilities.

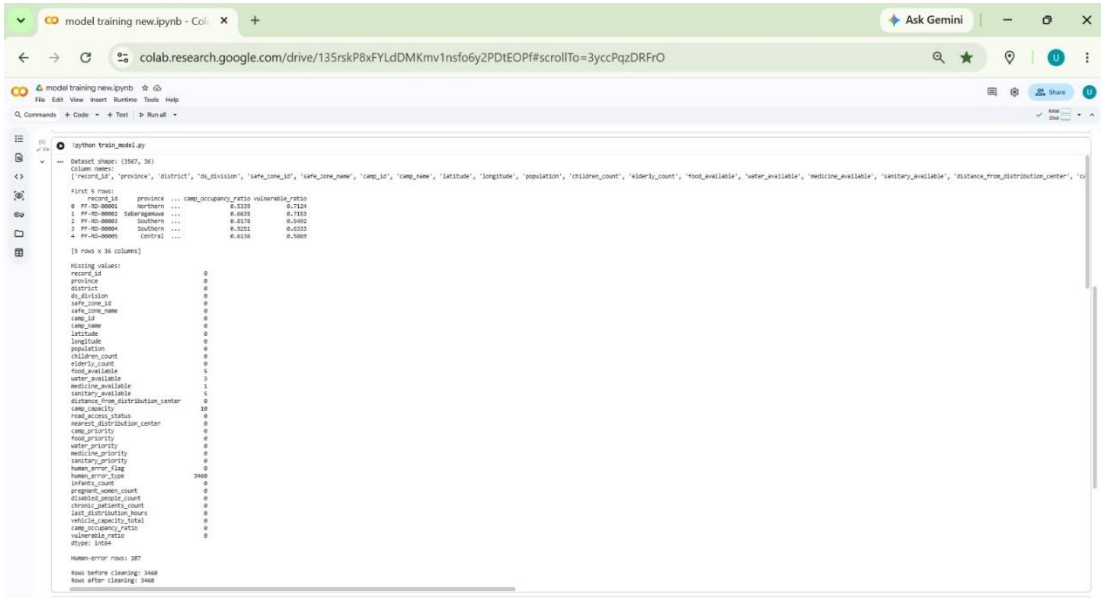
APPENDICES

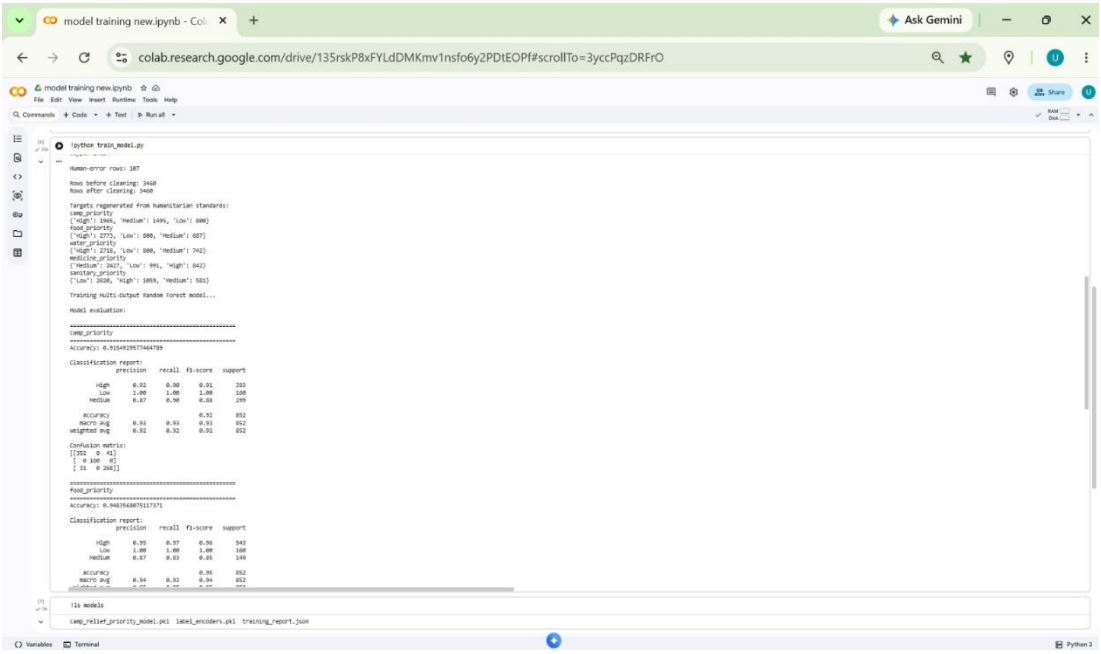
Appendix A Research instruments or extended results






model training new.ipynb - Col
Ask Gemini
colab.research.google.com/drive/135rskP8xFYLdDMKmv1nsfo6y2PDtEOPf#scrollTo=3yccPqzDRFrO
model training new.ipynb
!python train_model.py
Model evaluation:
camp_priority
Accuracy: 0.9154929577464789
Classification report:
precision recall f1-score support
High 0.92 0.90 0.91 393
Low 1.00 1.00 1.00 160
Medium 0.87 0.90 0.88 299
accuracy 0.92 852
macro avg 0.93 0.93 0.93 852
weighted avg 0.92 0.92 0.92 852
Confusion matrix:
[[352 0 41]
[0 160 0]
[31 0 268]]
food_priority
Accuracy: 0.9483568075117371
Classification report:
precision recall f1-score support
High 0.96 0.97 0.96 543
Low 1.00 1.00 1.00 160
Medium 0.87 0.83 0.85 149
accuracy 0.95 852
macro avg 0.94 0.93 0.94 852
weighted avg 0.95 0.95 0.95 852
Confusion matrix:
[[524 0 19]
[0 160 0]
[25 0 124]]
water_priority
Accuracy: 0.9624413145539906
Classification report:
precision recall f1-score support
High 0.98 0.96 0.97 545
Low 1.00 1.00 1.00 160
Medium 0.86 0.94 0.90 147
accuracy 0.96 852


model training new.ipynb - Col
Ask Gemini
colab.research.google.com/drive/135rskP8xFYLdDMKmv1nsfo6y2PDtEOPf#scrollTo=3yccPqzDRFrO
model training new.ipynb
!python train_model.py
Dataset shape: (3567, 36)
Column names:
First 5 rows:
record_id province ... camp_occupancy_ratio vulnerable_ratio
0 FF-RD-00001 Northern ... 0.5333 0.7124
1 FF-RD-00002 Sabaragamuwa ... 0.6635 0.7153
2 FF-RD-00003 Southern ... 0.6176 0.6491
3 FF-RD-00004 Southern ... 0.9251 0.6333
4 FF-RD-00005 Central ... 0.6136 0.5869
[5 rows x 36 columns]
Missing values:
record_id 0
province 0
district 0
ds_division 0
safe_zone_id 0
safe_zone_name 0
camp_id 0
camp_name 0
latitude 0
longitude 0
population 0
children_count 0
elderly_count 0
food_available 5
water_available 3
medicine_available 1
sanitary_available 5
distance_from_distribution_center 0
camp_capacity 10
road_access_status 0
nearest_distribution_center 0
camp_priority 0
food_priority 0
water_priority 0
medicine_priority 0
sanitary_priority 0
human_error_flag 0
human_error_type 3408
infants_count 0
pregnant_women_count 0
disabled_people_count 0
chronic_patients_count 0
last_distribution_hours 0
vehicle_capacity_total 0
camp_occupancy_ratio 0
vulnerable_ratio 0
dtype: int64
Human-error rows: 107
Rows before cleaning: 3468
Rows after cleaning: 3468


model training new.ipynb - Col
Ask Gemini
colab.research.google.com/drive/135rskP8xFYLdDMKmv1nsfo6y2PDtEOPf#scrollTo=3yccPqzDRFrO
model training new.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
!python train_model.py
Human-error rows: 107
Rows before cleaning: 3460
Rows after cleaning: 3460
Targets regenerated from humanitarian standards:
camp_priority
{'High': 1945, 'Medium': 1495, 'Low': 800}
food_priority
{'High': 2775, 'Low': 800, 'Medium': 687}
water_priority
{'High': 2718, 'Low': 800, 'Medium': 742}
medicine_priority
{'Medium': 2427, 'Low': 991, 'High': 842}
sanitary_priority
{'Low': 2620, 'High': 1059, 'Medium': 581}
Training Multi-Output Random Forest model...
Model evaluation:
==
camp_priority
==
Accuracy: 0.9154929577464789
Classification report:
precision recall f1-score support
High 0.92 0.90 0.91 393
Low 1.00 1.00 1.00 160
Medium 0.87 0.90 0.88 299
accuracy 0.91 852
macro avg 0.93 0.93 0.93 852
weighted avg 0.92 0.92 0.92 852
Confusion matrix:
[[352 0 41]
[0 160 0]
[31 0 268]]
==
food_priority
==
Accuracy: 0.9483568075117371
Classification report:
precision recall f1-score support
High 0.95 0.97 0.96 543
Low 1.00 1.00 1.00 160
Medium 0.87 0.83 0.85 149
accuracy 0.95 852
macro avg 0.94 0.93 0.94 852
!ls models
camp_relief_priority_model.pkl label_encoders.pkl training_report.json
Variables Terminal
Python 3

Appendix B: Academic Integrity and AI Use Disclosure

Academic Integrity Declaration

I declare that this individual thesis, titled **Post-Flood Rescue and Ration Distribution Management System**, has been prepared in accordance with the principles of academic integrity. All external sources, research publications, datasets, software tools, and other materials used in preparing this thesis must be appropriately acknowledged and referenced.

I acknowledge that I am responsible for the accuracy, originality, technical validity, and proper referencing of the submitted work. I understand that research findings, model performance results, and implementation details must accurately reflect the work undertaken.

AI Use Disclosure

Generative AI tools were used to assist with drafting, organizing, and refining selected sections of this thesis. The assistance included improving the clarity of technical explanations, structuring thesis content, and preparing descriptions of the Post-Flood Rescue and Ration Distribution Management System component.

The research topic, individual component scope, and project-specific technical context were based on the Flood Guard 360 project materials. AI-generated content was reviewed for relevance and consistency with the available project documentation. The student remains responsible for verifying technical statements, model evaluation results, references, implementation details, and conclusions before submission.

AI assistance was not treated as a substitute for implementation, experimental testing, independent analysis, or academic source verification. Any code, datasets, results, and claims included in the final thesis must accurately reflect the work performed by the student and the research team.

AI Tools Used and Their Purposes

Tool: ChatGPT (OpenAI)

Purpose of use:

- Assisting with the organization and drafting of thesis chapters and subsections.

- Improving the clarity, grammar, structure, and academic tone of written content.
- Drafting explanations of the Post-Flood Rescue and Ration Distribution Management System component, including its objectives, methodology, architecture, implementation, and integration within Flood Guard 360.
- Assisting with the presentation of model evaluation results and the discussion of research limitations.
- Preparing supporting text for the conclusion, future directions, glossary, and appendices.

ChatGPT was used as a writing and explanatory aid. It was not treated as an authoritative source of academic evidence, and AI-generated references or technical claims should be independently checked before submission.